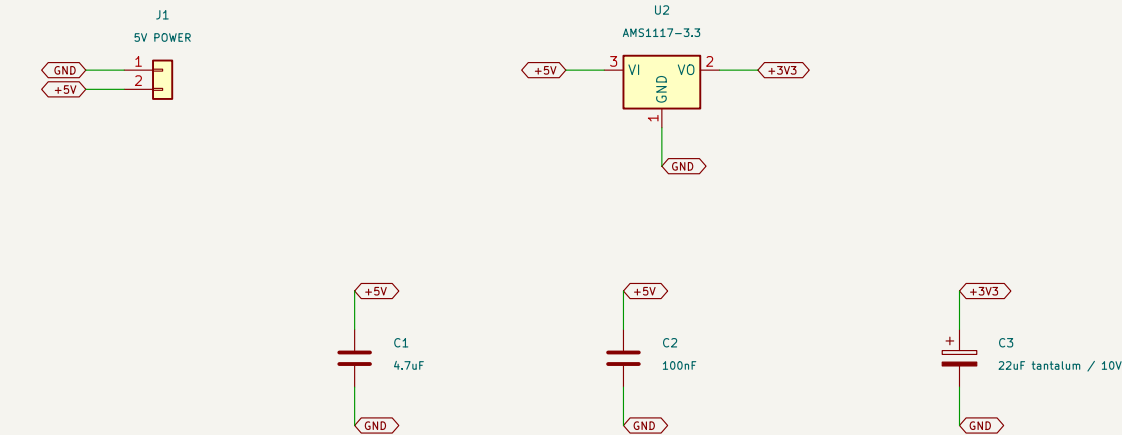


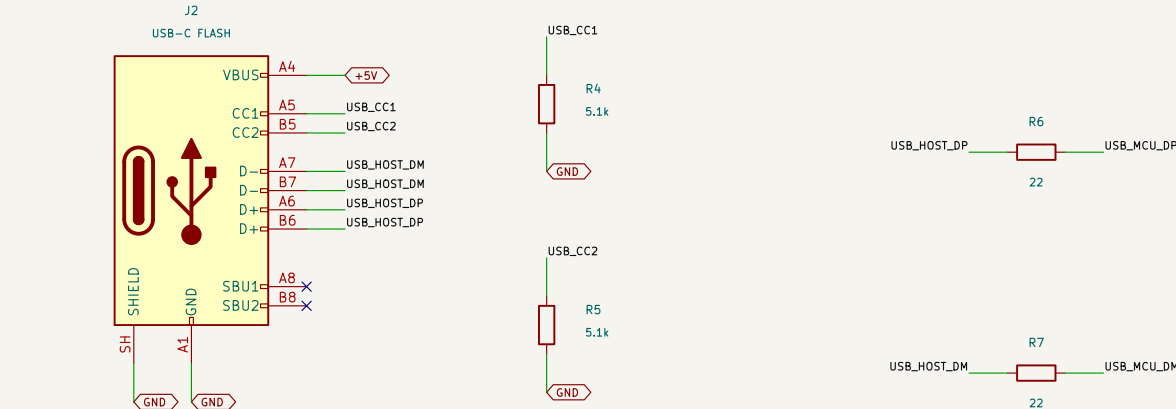
1. POWER – XH for production OR USB–C for flashing

2. ESP32–S3–MINI–1U–N8 – CAN only, no antenna fitted



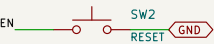
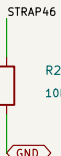
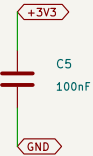
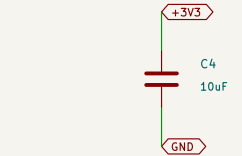
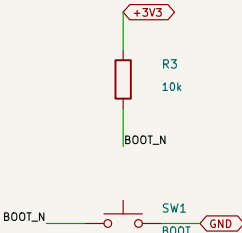
C3: solid tantalum for AMS1117 stability.
Provide copper heatsinking: 0.85W at 500mA; verify enclosure temperature.
USB VBUS and J1 feed +5V: NEVER connect both power sources.

3. USB–C – flashing and power; disconnect XH power before connecting USB

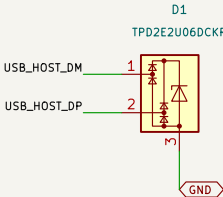


USB flashing: disconnect J1 power, then connect Mac USB–C.
Production: unplug USB–C, then connect J1 to RECOM 5V.
Both feed the same +5V rail; no power OR-ing or reverse blocking.

ESP32 provides its USB data pull-up internally. CC1/CC2: 5.1k to GND.
Route USB as a 90-ohm pair; D1 at J2, R6/R7 at U1.
C1: 4.7uF input bulk to reduce USB plug-in inrush.



GPIO45/46 low: 3.3V flash / download strap. GPIO3 unused.
Hold BOOT, tap RESET, release BOOT for ROM USB download.
MINI–1U body: 15.4 x 15.4mm. Wi-Fi and BLE disabled in firmware.



Sensors interfaces

File: sensors_interfaces.kicad_sch

CAN-only / USB flash power OR external 5V

CoffeeFlow

Sheet: /

File: sensors.kicad_sch

Title: Sensors – power, ESP32–S3 and USB flashing

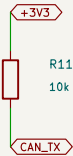
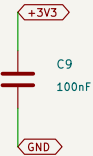
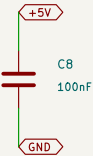
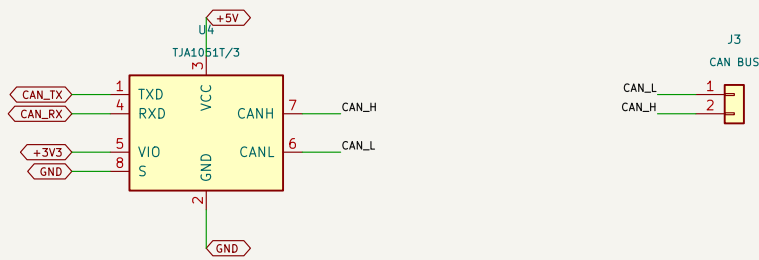
Size: A3 Date: 2026–10–01

Rev: A

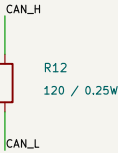
KiCad E.D.A. 10.0.6

Id: 1/2

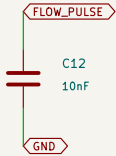
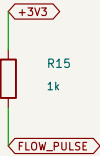
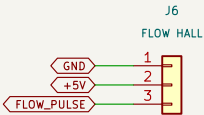
4. CAN – TJA1051T/3, 5V bus driver / 3.3V logic



S/SLNT hard-wired low. R11 keeps TX recessive during reset. R12 permanently terminates this bus-end node (120 ohms). J3 has no GND pin: power GND must run alongside CAN pair.

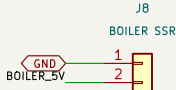
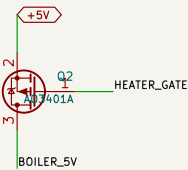
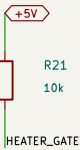
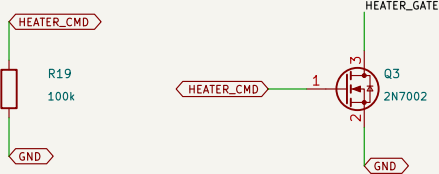


6. DIGMESA – open collector, existing RC filter



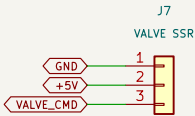
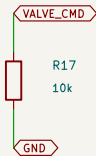
GPIO41 input; internal pull-up disabled. 1k x 10nF = 10us. Digmesa powered from 5V; signal pulled to 3.3V only. Pin order: GND brown / 5V green / SIGNAL yellow. RC discharge speed also depends on sensor/cable impedance.

8. BOILER SSR – powered 5V output, HIGH command = ON

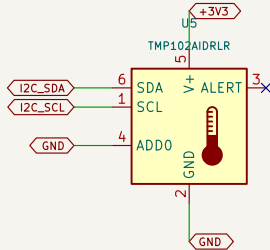
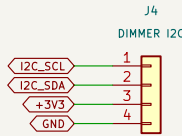
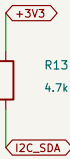


GPIO36, 3.3V signal / 5V VCC; default LOW.

7. VALVE SSR – 3.3V signal / 5V supply



5. SHARED I2C – 3.3V, Grove/HY2.0 cable order



Grove opening up: left-to-right GND / 3V3 / SDA / SCL. Numbered pads: 1=SCL yellow, 2=SDA white, 3=3V3 red, 4=GND black. Addresses: dimmer 0x50 / XDB401 0x7F / TMP102 0x48. TMP102 ADD0 grounded; ALERT unused. Mount away from LDO heat. Pull-ups always fitted: 2.35k effective with XDB401 attached. 100kHz initial bus speed; remote modules provide local supply decoupling.

J8: SSR (-) -> pin1 GND; SSR (+) -> pin2 switched 5V. Q3 pulls Q2 gate LOW; Q2 supplies the SSR current. R19/R21/R22 ensure off with GPIO floating or 3V3 absent. Shared GND. Gate pull-up current about 0.5mA when ON; load <=100mA; no mains.

CAN-only / USB flash power OR external 5V

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Sheet: /Sensors interfaces/

File: sensors_interfaces.kicad_sch

Title: Sensors – CAN, I2C, flowmeter and SSR outputs

Size: A3

Date: 2026-10-01

Rev: A

KiCad E.D.A. 10.0.6

Id: 2/2